

Name: _____

Date: _____

NUMBER OPERATIONS, (MULTI-STEP CALCULATIONS)

LO: I can solve problems that require more than one step, choosing the correct operations.

Multi-step problems

A **multi-step calculation** requires you to perform two or more **operations** in sequence to reach the answer.

Strategy: Read the problem carefully. Identify what you need to find. Break it into smaller steps. Use BIDMAS when needed.

Approach

●	Read Plan Calculate Check	4 steps
●	Break complex problems into parts	simplify
●	Use BIDMAS for order of operations	always
●	Just do operations left to right	must use BIDMAS
●	Only use one operation per problem	multi-step needs more

Key idea

Break the problem into steps. Solve each step in order. Check your answer makes sense.

Step 1

Step 2

Answer

One step at a time

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – shopping problem

Buy 3 shirts at £12 each and 2 pairs of socks at £3.50. Total cost?

$$\text{Shirts: } 3 \times 12 = £36$$

$$\text{Socks: } 2 \times 3.50 = £7$$

$$\text{Total: } 36 + 7 = £43$$

= £43

Calculate each item separately, then add.

Example 2 – with change

You pay for a £7.80 item with a £20 note. How many 50p coins in your change?

$$\text{Change: } 20 - 7.80 = £12.20$$

$$\begin{aligned} \text{Divide by 0.50: } 12.20 \div 0.50 &= 24.4 \\ &= 24 \text{ complete 50p coins (with 20p left)} \end{aligned}$$

= 24 coins

Find the change first, then divide by the coin value.

Example 3 – distance/speed

A car travels at 60 mph for 2.5 hours then 40 mph for 1.5 hours. Total distance?

$$\text{Part 1: } 60 \times 2.5 = 150 \text{ miles}$$

$$\text{Part 2: } 40 \times 1.5 = 60 \text{ miles}$$

$$\text{Total: } 150 + 60 = 210 \text{ miles}$$

= 210 miles

Distance = speed $\times$ time for each section, then add.

Example 4 – profit calculation

Buy 50 items at £3 each. Sell 40 at £5 and 10 at £2. Total profit?

$$\text{Cost: } 50 \times 3 = £150$$

$$\begin{aligned} \text{Revenue: } (40 \times 5) + (10 \times 2) &= 200 + \\ &20 = £220 \end{aligned}$$

$$\text{Profit: } 220 - 150 = £70$$

= £70 profit

Profit = revenue - cost.

YOUR TURN – PRACTICE (deliberate practice)

1. Two-step calculations

Solve each two-step problem.

- a) $4 \times 7 + 3 \times 5$
- b) $100 - 6 \times 12$
- c) $(15 + 25) \div 8$
- d) $3 \times (20 - 8) + 4$

2. Money problems

Show your working.

- a) 5 items at £4.50. Change from £30?
- b) 3 meals at £8.95 + 2 drinks at £2.30. Total?
- c) Earn £9.50/hour for 8 hours. Spend £45. Left?
- d) Buy 4 at £6.25. Pay with 2 × £20 notes. Change?

3. Rate problems

Calculate step by step.

- a) 60 mph for 3 hours then 45 mph for 2 hours. Total miles?
- b) Type 50 words/min for 12 min. How many words?
- c) Fill 8 litres/min for 5 min, drain 3 litres/min for 4 min. Net?
- d) Earn £12/hr weekday, £18/hr weekend. Work 30 + 8 hrs. Total?

4. Real-life multi-step

Read carefully and solve.

- a) Bus fare £2.40 each way, 5 days/week. Weekly cost?
- b) Recipe for 4: needs 300g flour. How much for 10 people?
- c) Pool: 25m. Swim 12 lengths. Total in km?
- d) 6 eggs/box, need 50 eggs. How many boxes to buy?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Multi-step problems need more than one operation ☐

Correct answer: _____

Reason: _____

b) Always add all the numbers in a word problem ☐

Correct answer: _____

Reason: _____

c) Break problems into smaller steps ☐

Correct answer: _____

Reason: _____

d) The operations are always stated explicitly ☐

Correct answer: _____

Reason: _____

e) Check your answer makes sense in context ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A school trip costs **£15 per student** plus a fixed **£200** for the coach. There are **32 students**. They sell cakes for **£1.50 each** to raise money. How many cakes must they sell to cover the full cost?

Expression: _____

Simplified: _____

b) Paint covers **12 m² per litre**. A room has 4 walls each **3 m × 2.5 m**, minus a door (**2 m × 0.8 m**) and a window (**1.5 m × 1 m**). How many litres of paint are needed? (Round up to whole litres.)

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

